

Reshma Rajendran

Machine Learning Engineer | 2+ years ML experience (audio, vision, NLP, LLM) | MTech IIT Tirupati

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Machine Learning Engineer with experience in pose estimation for real-time hand gesture recognition and a strong background in ML model development and optimization. Skilled in Python, PyTorch, TensorFlow, OpenCV. Experienced across CV, NLP, audio detection and TTS projects, with a focus on building efficient, low-latency AI systems.

EXPERIENCE

Aiverbalyze technologies | Machine learning Engineer

May 2025–present | India

Fine-tuned a Llama based TTS model, to generate highly expressive, natural-sounding speech in multiple Indian languages.

- Engineered a “The voice” production-grade multilingual TTS system trained on 2.2M+ audio-text pairs, achieving low training loss of 3.43 and ~500ms streaming latency, ensuring seamless interaction with the TTS system.
- Optimized training with Unsloth (LoRA) and Axolotl, reducing compute cost, training time, and memory footprint by 50% while maintaining model quality; deployed models using ONNX for efficient inference.

Abili-Tech System | Machine learning Engineer

May 2024–April 2025 | BC, Canada

Developed a real-time sign language translation web app using Streamlit, deployed on Hugging Face Spaces.

- Implemented a Random Forest model achieving ~75% translation accuracy for gesture-to-text conversion.
- Used OpenCV for real-time video capture and optimized data preprocessing, improving responsiveness and reliability.

NXP Semiconductors | Machine learning Intern

Jun 2022–Jun 2023 | Bangalore, India

Collaborated within NXP Semiconductors' R&D team, gaining expertise in audio classification and signal analysis.

- Designed and implemented audio detection system with specific audio features leveraging dominant frequency and high SNR, achieving 22% Equal Error Rate over baseline features with lower computational complexity.
- Applied Gaussian Mixture Model (GMM) for multi-class audio inference, enhanced feature quality with custom filter bank and spectrum thresholding and proposed temporal variation analysis for future improvements.

Tata Consultancy Services | Assistant System Engineer

Aug 2017–April 2019 | Chennai, India

- Packaged, tested, and deployed applications via SCCM during a Windows 10 migration for 10,000+ users
- Managed Hyper-V hosts and internal lab infrastructure for demos and testing, while developing PowerShell and Azure automation scripts to streamline SCCM tasks, reducing manual effort and improving operational efficiency.

PROJECTS

Anomaly Detection on MVTec Dataset – Explored multiple approaches for anomaly detection including HOG+SVM and autoencoders, which underperformed on certain categories (AUC ~0.36–0.74). Developed a ResNet-based feature extraction pipeline with a neural network classifier, achieving a significant performance boost to 0.86 AUC.

Real-time Bicep Curl Counter – Built a real-time pose estimation system using MediaPipe and OpenCV to track keypoints, calculate elbow angles, and count bicep curls. Developed a live feedback interface displaying angles, stage (up/down), and rep count. Conducted a pose estimation comparison with YOLOv7 on recorded videos to evaluate accuracy, robustness, and potential improvements.

End-to-End Signboard Translation System – Developed an end-to-end signboard translation pipeline with YOLOv8n-based text detection (96.7% mAP@50), CNN-RNN OCR (CER 0.36→0.12, WER 0.35→0.21), and Seq2Seq translation (BLEU 57.9, ROUGE 0.18, Perplexity 0.20). Optimized performance using preprocessing, augmentation, and training strategies, while tracking experiments with Weights & Biases.

SKILLS

Languages/libraries: Python (pandas, NumPy, Scikit-Learn, NLTK, OpenCV, PyTorch, TensorFlow), MATLAB, SQL

Computer Vision: Image preprocessing, feature extraction, classification, segmentation, object detection, pose estimation

NLP: Text preprocessing, probabilistic models, sequence models, attention models, transformers

EDUCATION

Indian Institute of Technology (IIT) Tirupati

Aug 2021–May 2023 | Tirupati, India

Master of Technology (MTech) in Electrical Engineering | GPA: 8.73/10

- Specialization: Signal Processing and Communication
- Relevant Courses: Deep Learning for Computer Vision, Machine Learning for Image Processing

Rajalakshmi Engineering College

Aug 2013–May 2017 | Chennai, India

Bachelor of Engineering (BE) in Electronics and Communication Engineering | GPA: 8.46/10

CERTIFICATIONS

Deep Learning online course | One Fourth LABS (IIT MADRAS)

Mastering OpenCV with Python | OpenCV University

AWS Certified Cloud Practitioner | AWS